# 1 Important Inequalities

The proofs of the error bound formulas all use the *Triangle Inequality*

$$\left|\sum_{i=1}^{n} x_i\right| \leq \sum_{i=1}^{n} |x_i|,$$

which you proved in AEP #1, as well as the following similar-looking inequality for integrals.

**Proposition 1.1** (Triangle inequality for integrals)**.** *If $f : [a,b] \to \mathbb{R}$ is integrable then so is $|f|$ and*

$$\left|\int_a^b f(x)dx\right| \leq \int_a^b |f(x)|dx.$$

1. Assume the first statement in Proposition 1.1 and prove the second.
(*Hint*: $-|f(x)| \leq f(x) \leq |f(x)|$.)

# 2 Simpson's Rule

**Theorem 2.1** (Error bound for Simpson's Rule)**.** *Let $f(x)$ be a continuous function over $[a,b]$ having a fourth derivative $f^{(4)}(x)$ over this interval. If $M = \max\{|f^{(4)}(x)|\}_{x\in[a,b]}$, then the upper bound for the error in approximating $\int_a^b f(x)dx$ using Simpson's Rule is*

$$E_n^S(f) \leq \frac{M(b-a)^5}{180n^4}.$$

*Proof.* (a) Consider the piecewise function [1]

$$p(x) = \begin{cases} (x-x_{2i-2})^3(x-\frac{x_{2i-2}+2x_{2i}}{3}), & x_{2i-2} \leq x \leq x_{2i-1}, \\ (x-x_{2i})^3(x-\frac{2x_{2i-2}+x_{2i}}{3}), & x_{2i-1} \leq x \leq x_{2i}. \end{cases}$$

(i) Calculate and simplify $p^{(n)}(x)$ for $n = 1,2,3,4$. Use the simplified expression for $p'(x)$ to calculate $p''(x)$, and so on. (If this becomes too time-consuming, you may calculate these using *Mathematica* and write the simplified answers here.)

(ii) Verify that $p(x_{2i-2}) = p'(x_{2i-2}) = p''(x_{2i-2}) = p(x_{2i}) = p'(x_{2i}) = p''(x_{2i}) = 0$.

[1] This function is constructed to have the properties listed below, so that certain terms vanish upon the repeated integration by parts we will perform later. We could construct this function from scratch, but it is a lot of algebra and would be very time-consuming.

(iii) Show that $p^{(n)}(x)$ for $n = 0, 1, 2$ is continuous on $[x_{2i-2}, x_{2i}]$. (*Hint*: $x_{2i-1} = \frac{x_{2i-2}+x_{2i}}{2}$.)

(iv) Show that $p^{(3)}(x)$ is continuous on $[x_{2i-2}, x_{2i-1}]$ and $[x_{2i-1}, x_{2i}]$, but has a jump discontinuity at $x_{2i-1}$.

(v) Calculate $p^{(3)}(x_{2i-2})$ and $p^{(3)}(x_{2i})$.

(b) Consider now the integral $\int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx$. Integrating by parts repeatedly,

$$\int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx = f^{(3)}(x)p(x)\Big|_{x_{2i-2}}^{x_{2i}} - f^{(2)}(x)p'(x)\Big|_{x_{2i-2}}^{x_{2i}} + f'(x)p''(x)\Big|_{x_{2i-2}}^{x_{2i}} - \int_{x_{2i-2}}^{x_{2i}} f'(x)p^{(3)}(x)dx.$$

Since $f(x)p^{(3)}(x)$ is continuous except for a single jump discontinuity, it is integrable. In order to use the Fundamental Theorem of Calculus to calculate the integral, we write it as

$$\int_{x_{2i-2}}^{x_{2i}} f'(x)p^{(3)}(x)dx = \int_{x_{2i-2}}^{x_{2i-1}} f'(x)p^{(3)}(x)dx + \int_{x_{2i-1}}^{x_{2i}} f'(x)p^{(3)}(x)dx,$$

so that each integrand is continuous over the corresponding subinterval. Therefore

$$\begin{aligned}\int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx = {} & f^{(3)}(x)p(x)\Big|_{x_{2i-2}}^{x_{2i}} - f^{(2)}(x)p'(x)\Big|_{x_{2i-2}}^{x_{2i}} + f'(x)p''(x)\Big|_{x_{2i-2}}^{x_{2i}} \\ & - \int_{x_{2i-2}}^{x_{2i-1}} f'(x)p^{(3)}(x)dx - \int_{x_{2i-1}}^{x_{2i}} f'(x)p^{(3)}(x)dx.\end{aligned}$$

Integrating each of the two remaining integrals by parts once more gives

$$\begin{aligned}\int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx = {} & f^{(3)}(x)p(x)\Big|_{x_{2i-2}}^{x_{2i}} - f^{(2)}(x)p'(x)\Big|_{x_{2i-2}}^{x_{2i}} + f'(x)p''(x)\Big|_{x_{2i-2}}^{x_{2i}} \\ & - f(x)p^{(3)}(x)\Big|_{x_{2i-2}}^{x_{2i-1}} - f(x)p^{(3)}(x)\Big|_{x_{2i-1}}^{x_{2i}} + \int_{x_{2i-2}}^{x_{2i-1}} f(x)p^{(4)}(x)dx \\ & + \int_{x_{2i-1}}^{x_{2i}} f(x)p^{(4)}(x)dx.\end{aligned}$$

By Problem 7(f), $f(x)p^{(4)}(x)$ is defined and is continuous at all points of $[x_{2i-2}, x_{2i}]$ except for $x_{2i-1}$, where it is undefined. Since changing the value of a continuous function at a single point does not change the integral, we have

$$\int_{x_{2i-2}}^{x_{2i-1}} f(x)p^{(4)}(x)dx + \int_{x_{2i-1}}^{x_{2i}} f(x)p^{(4)}(x)dx = \int_{x_{2i-2}}^{x_{2i}} f(x)p^{(4)}(x)dx$$

and therefore

$$\begin{aligned}\int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx = {} & f^{(3)}(x)p(x)\Big|_{x_{2i-2}}^{x_{2i}} - f^{(2)}(x)p'(x)\Big|_{x_{2i-2}}^{x_{2i}} + f'(x)p''(x)\Big|_{x_{2i-2}}^{x_{2i}} \\ & - f(x)p^{(3)}(x)\Big|_{x_{2i-2}}^{x_{2i-1}} - f(x)p^{(3)}(x)\Big|_{x_{2i-1}}^{x_{2i}} + \int_{x_{2i-2}}^{x_{2i}} f(x)p^{(4)}(x)dx \\ = {} & f^{(3)}(x)p(x)\Big|_{x_{2i-2}}^{x_{2i}} - f^{(2)}(x)p'(x)\Big|_{x_{2i-2}}^{x_{2i}} + f'(x)p''(x)\Big|_{x_{2i-2}}^{x_{2i}} \\ & - f(x)p^{(3)}(x)\Big|_{x_{2i-2}}^{x_{2i-1}} - f(x)p^{(3)}(x)\Big|_{x_{2i-1}}^{x_{2i}} + 24\int_{x_{2i-2}}^{x_{2i}} f(x)dx.\end{aligned}$$

(c) Use your answers from Problem 7(b)-(f) to show that this implies

$$24\int_{x_{2i-2}}^{x_{2i}} f(x)dx = -f(x)p^{(3)}(x)\Big|_{x_{2i-2}}^{x_{2i-1}} - f(x)p^{(3)}(x)\Big|_{x_{2i-1}}^{x_{2i}} + \int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx$$

and therefore

$$24\int_{x_{2i-2}}^{x_{2i}} f(x)dx = -f(x_{2i-1})p^{(3)}(x_{2i-1}^{-}) + f(x_{2i-2})p^{(3)}(x_{2i-2}) - f(x_{2i})p^{(3)}(x_{2i}) + f(x_{2i-1})p^{(3)}(x_{2i-1}^{+}) \int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx.$$

(d) Again, use your answers from 7(b)-(f) to show that this expression simplifies to

$$\int_{x_{2i-2}}^{x_{2i}} f(x)dx = \frac{x_{2i}-x_{2i-2}}{6}\left(f(x_{2i-2}) + 4f(x_{2i-1}) + f(x_{2i})\right) + \frac{1}{24}\int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx.$$

It follows that

$$\int_a^b f(x)dx = \sum_{i=1}^{n/2} \int_{x_{2i-2}}^{x_{2i}} f(x)dx$$
$$= \sum_{i=1}^{n/2} \frac{x_{2i} - x_{2i-2}}{6} \left(f(x_{2i-2}) + 4f(x_{2i-1}) + f(x_{2i})\right) + \frac{1}{24} \sum_{i=1}^{n/2} \int_{x_{2i-2}}^{x_{2i}} f^{(4)}(x)p(x)dx.$$

(e) Show that the expression above implies that $|E_n^S(f)| \leq \frac{M}{24} \sum_{i=1}^{n/2} \int_{x_{2i-2}}^{x_{2i}} |p(x)|dx$.

Using the definition of $p(x)$, we can write

$$\begin{aligned}
&\frac{M}{24}\sum_{i=1}^{n/2}\int_{x_{2i-2}}^{x_{2i}}|p(x)|dx \\
&=\frac{M}{24}\sum_{i=1}^{n/2}\left[\int_{x_{2i-2}}^{x_{2i-1}}\left|(x-x_{2i-2})^3(x-\frac{x_{2i-2}}{3}-\frac{2x_{2i}}{3})\right|dx+\int_{x_{2i-1}}^{x_{2i}}\left|(x-x_{2i})^3(x-\frac{2x_{2i-2}}{3}-\frac{x_{2i}}{3})\right|dx\right] \\
&=\frac{M}{24}\sum_{i=1}^{n/2}\left[\int_{x_{2i-2}}^{x_{2i-1}}|x-x_{2i-2}|^3\left|x-\frac{x_{2i-2}}{3}-\frac{2x_{2i}}{3}\right|dx+\int_{x_{2i-1}}^{x_{2i}}|x-x_{2i}|^3\left|x-\frac{2x_{2i-2}}{3}-\frac{x_{2i}}{3}\right|dx\right] \\
&=\frac{M}{24}\sum_{i=1}^{n/2}\left[\int_{x_{2i-2}}^{x_{2i-1}}(x-x_{2i-2})^3\left|x-\frac{x_{2i-2}}{3}-\frac{2x_{2i}}{3}\right|dx+\int_{x_{2i-1}}^{x_{2i}}(x_{2i}-x)^3\left|x-\frac{2x_{2i-2}}{3}-\frac{x_{2i}}{3}\right|dx\right].
\end{aligned}$$

(a) Show that for $x_{2i-2}\le x\le x_{2i-1}$, $x-\frac{x_{2i-2}}{3}-\frac{2x_{2i}}{3}<0$, and therefore $\left|x-\frac{x_{2i-2}}{3}-\frac{2x_{2i}}{3}\right|=\frac{x_{2i-2}}{3}+\frac{2x_{2i}}{3}-x$. (*Hint*: Recall that $x_{2i-1}=\frac{x_{2i-2}+x_{2i}}{2}$.)

(b) Show that for $x_{2i-1}\le x\le x_{2i}$, $x-\frac{2x_{2i-2}}{3}-\frac{x_{2i}}{3}>0$ and therefore $\left|x-\frac{2x_{2i-2}}{3}-\frac{x_{2i}}{3}\right|=x-\frac{2x_{2i-2}}{3}-\frac{x_{2i}}{3}$.

It follows from the previous problem that

$$\int_{x_{2i-2}}^{x_{2i-1}} (x-x_{2i-2})^3 \left| x - \frac{x_{2i-2}}{3} - \frac{2x_{2i}}{3} \right| dx + \int_{x_{2i-1}}^{x_{2i}} (x_{2i}-x)^3 \left| x - \frac{2x_{2i-2}}{3} - \frac{x_{2i}}{3} \right| dx$$
$$= \int_{x_{2i-2}}^{x_{2i-1}} (x-x_{2i-2})^3 \left( \frac{x_{2i-2}}{3} + \frac{2x_{2i}}{3} - x \right) dx + \int_{x_{2i-1}}^{x_{2i}} (x_{2i}-x)^3 \left( x - \frac{2x_{2i-2}}{3} - \frac{x_{2i}}{3} \right) dx$$

(f) Show that

$$\int_{x_{2i-2}}^{x_{2i-1}} (x-x_{2i-2})^3 \left( \frac{x_{2i-2}}{3} + \frac{2x_{2i}}{3} - x \right) dx = \int_{x_{2i-1}}^{x_{2i}} (x_{2i}-x)^3 \left( x - \frac{2x_{2i-2}}{3} - \frac{x_{2i}}{3} \right) dx$$

and therefore

$$\int_{x_{2i-2}}^{x_{2i-1}} (x-x_{2i-2})^3 \left( \frac{x_{2i-2}}{3} + \frac{2x_{2i}}{3} - x \right) dx + \int_{x_{2i-1}}^{x_{2i}} (x_{2i}-x)^3 \left( x - \frac{2x_{2i-2}}{3} - \frac{x_{2i}}{3} \right) dx$$
$$= 2\int_{x_{2i-2}}^{x_{2i-1}} (x-x_{2i-2})^3 \left( \frac{x_{2i-2}}{3} + \frac{2x_{2i}}{3} - x \right) dx$$

by making the substitution $u = x_{2i-2} + x_{2i} - x$ in $\int_{x_{2i-1}}^{x_{2i}} (x_{2i}-x)^3 \left( x - \frac{2x_{2i-2}}{3} - \frac{x_{2i}}{3} \right) dx$.

(g) Show that $\frac{x_{2i-2}}{3}+\frac{2x_{2i}}{3}-x=\frac{2}{3}(x_{2i}-x_{2i-2})-(x-x_{2i-2})$ and therefore

$$\int_{x_{2i-2}}^{x_{2i-1}}(x-x_{2i-2})^3\left(\frac{x_{2i-2}}{3}+\frac{2x_{2i}}{3}-x\right)dx=\int_{x_{2i-2}}^{x_{2i-1}}(x-x_{2i-2})^3\left(\frac{2}{3}(x_{2i}-x_{2i-2})-(x-x_{2i-2})\right)dx.$$

(*Hint*: $\frac{x_{2i-2}}{3}+\frac{2x_{2i}}{3}-x=\frac{x_{2i-2}}{3}+\frac{2x_{2i}}{3}-x+x_{2i-2}-\frac{3x_{2i-2}}{3}$)

(h) Show that $\int_{x_{2i-2}}^{x_{2i-1}}(x-x_{2i-2})^3\left(\frac{2}{3}(x_{2i}-x_{2i-2})-(x-x_{2i-2})\right)dx=\frac{1}{240}(x_{2i}-x_{2i-2})^5$.

(i) Put everything together to complete the proof.

□